



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE 'B'

Course Code & Title: CS3491 Artificial Intelligence and Machine Learning

Name of the Faculty member (s): Dr.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit V / Deep Networks**

- **Course Outcome: CO5**

- **Topic Learning Outcome: 5d**

- **Activity Chosen: Crossword Puzzle**

- **Justification:**

Using a crossword puzzle as an active learning technique for teaching neural network concepts encourages active recall, reinforces key terminology, and promotes deeper understanding through conceptual connections. Crossword puzzles also help identify misconceptions and knowledge gaps, making them a valuable diagnostic tool. This approach transforms passive learning into an interactive experience, fostering retention and meaningful engagement with complex deep learning concepts like optimization and hyper parameter tuning.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**

- The deep neural network concepts are taught to the students. At the end of the session the crossword puzzle activity was conducted.
- A crossword puzzle sheet consists of 10 questions was given to the student's team of 2 members to improve peer learning collaboration.
- They were asked to complete the puzzle within the duration of 10 minutes.
- After the completion, the answers were reviewed. The terms given in the puzzle are explained clearly and the doubts are clarified.
- The student's involvement in the activity is shown in Fig.1 and the sample answers given by the students is shown in Fig.2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	3	2	2	3	1	2	2	1	2	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO3
	3	2	1	1	3

Justification for correlation	The students has the knowledge of deep neural network.	The students identify the methods for training and optimization	The students work as a team to share their solution.	The students communicate the solution to others.	The concept of neural network is essential for solving real world problems using AI
--------------------------------------	--	---	--	--	---

• **Images / Screenshot of the practice:**



Fig 1. Student Involvement

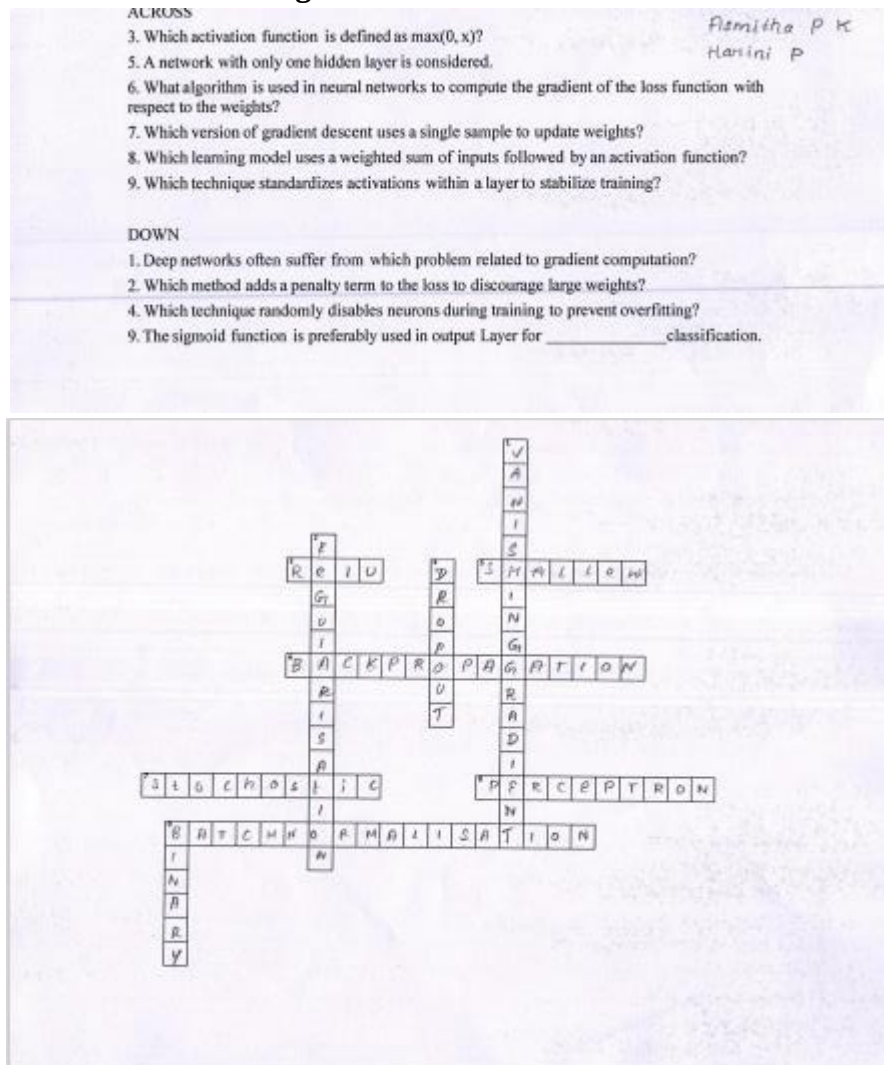


Fig 2. Solved Crossword Puzzle

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported increased engagement and better retention of technical terms due to the gamified format.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students understanding level of the concepts can be identified easily.
- It engages students in a fun, low-stakes activity that supports collaborative learning and peer instruction.

- ❖ ***Challenges faced in implementation:***

- Some of the students who are not familiar may struggle to find the answer from the given clues need more support to complete.

References:

- <https://puzzlemaker.discoveryeducation.com/criss-cross>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/RV-CWP-CS8601_MC_UIV.pdf